IPL Sponsorship Value Intelligence Dashboard using Business Intelligence and Machine Learning

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Abstract

The rapid growth of digital media and fan interaction has significantly transformed sports sponsorship evaluation in modern cricket leagues such as the Indian Premier League (IPL). Traditional sponsorship assessment methods primarily focus on television exposure and attendance, while modern sponsorship effectiveness increasingly depends on fan engagement, online popularity, and digital interaction. This project develops an IPL Sponsorship Value Intelligence Dashboard using Business Intelligence (BI) and Machine Learning (ML) techniques to analyze and predict sponsorship value across IPL teams and matches.

The study integrates multiple data sources, including IPL match data, Google Trends popularity data, YouTube engagement metrics, and stadium attendance proxies. A Sponsorship Value Index (SVI) was developed using normalized indicators such as audience exposure, digital engagement, and popularity scores. The dashboard was implemented in Power BI to provide interactive visualizations for sponsorship analysis and decision-making.

Additionally, a Machine Learning model was developed to predict future sponsorship value for IPL matches using historical engagement and popularity patterns. The predictive model demonstrated strong performance, indicating that fan engagement and audience exposure significantly influence sponsorship opportunities.

The developed system provides sports organizations and sponsors with data-driven insights to identify high-value sponsorship opportunities, evaluate fan engagement patterns, and support strategic sponsorship decisions in IPL cricket.

1. Introduction

The Indian Premier League (IPL) is one of the world's most commercially successful cricket leagues, attracting millions of viewers, sponsors, and digital audiences globally. Sponsorship plays a critical role in the financial ecosystem of IPL teams and franchises. With the rise of digital platforms and social media, sponsorship evaluation has evolved beyond traditional television reach and stadium attendance metrics.

Modern sponsors increasingly rely on digital engagement, fan interaction, online popularity, and audience exposure to assess sponsorship effectiveness. Consequently, sports organizations require intelligent analytical systems capable of integrating multiple data sources and generating actionable insights for sponsorship decision-making.

Business Intelligence (BI) and Machine Learning (ML) technologies provide opportunities to transform raw sports data into strategic sponsorship intelligence. BI dashboards enable interactive visualization of sponsorship performance, while ML models can forecast future sponsorship potential using historical trends and engagement patterns.

This project aims to develop an IPL Sponsorship Value Intelligence Dashboard that integrates IPL match data, Google Trends popularity data, YouTube engagement metrics, and audience exposure indicators to analyze and predict sponsorship value across IPL teams and matches.

The major objectives of the study are:

1. To analyze fan engagement and popularity patterns across IPL teams.
2. To develop a Sponsorship Value Index (SVI) using engagement and exposure metrics.
3. To build an interactive Power BI dashboard for sponsorship intelligence.
4. To apply Machine Learning techniques for predicting future sponsorship value.

The proposed system supports sponsors, franchises, and sports analysts in identifying high-value sponsorship opportunities using data-driven insights.

2. Literature Review

Sports sponsorship analytics has gained increasing attention due to the commercialization of professional sports leagues and the growth of digital media platforms. Researchers have emphasized the importance of fan engagement, audience exposure, and social media interaction in evaluating sponsorship effectiveness.

Previous studies indicate that sponsorship value is strongly influenced by audience reach, team popularity, and fan interaction levels. Traditional sponsorship evaluation primarily focused on television ratings and stadium attendance; however, digital engagement metrics such as likes, comments, and online searches have become increasingly significant in modern sports marketing.

Several researchers have explored the application of Business Intelligence in sports analytics. BI systems enable organizations to integrate multiple data sources, generate interactive dashboards, and support strategic decision-making. Sports organizations increasingly use dashboards to monitor fan engagement, player performance, sponsorship effectiveness, and audience behavior.

Machine Learning techniques have also been applied in sports analytics for predictive modeling, trend forecasting, and performance evaluation. Studies suggest that predictive analytics can support sponsorship planning by identifying future high-value matches and teams based on historical engagement patterns.

Google Trends has been widely used as a proxy for public interest and online popularity in sports analytics research. Similarly, YouTube engagement metrics such as views, likes, and comments are increasingly used to evaluate fan interaction and digital influence.

Despite growing research in sports analytics, limited studies integrate social media engagement, popularity analytics, and sponsorship intelligence within a unified BI and ML framework for IPL sponsorship evaluation. This project addresses this gap by developing an integrated Sponsorship Value Intelligence Dashboard using multi-source analytics and predictive modeling.

3. Methodology

3.1 Data Collection

The study integrates data from multiple sources:

Data Source	Purpose
IPL Match Dataset	Match statistics and venue details
Google Trends	Team popularity analysis
YouTube API	Fan engagement analysis
Stadium Capacity Data	Audience exposure proxy

The IPL match dataset includes information such as match date, teams, venue, winner, and player of the match. Google Trends data was collected for selected IPL teams to measure online popularity and public interest. YouTube API data was used to extract views, likes, and comments related to IPL team highlights and promotional content.

3.2 Data Preprocessing

The collected data underwent preprocessing steps including:

- Handling missing values
- Standardizing team names
- Feature normalization
- Data merging and integration

Venue names were cleaned and mapped with stadium capacity values to estimate audience exposure using attendance proxies.

3.3 Fan Engagement Analysis

YouTube engagement metrics were used to calculate team-level engagement using the following formula:

$$Engagement = (Likes + Comments)/(Views)$$

Google Trends popularity scores were integrated with engagement data to evaluate overall team popularity and fan interaction.

3.4 Sponsorship Value Index (SVI)

A Sponsorship Value Index (SVI) was developed using normalized indicators:

$$SVI = 0.4(Audience\ Exposure) + 0.3(Popularity\ Score) + 0.3(Match\ Engagement)$$

Where:

- Audience Exposure = Stadium attendance proxy
- Popularity Score = Google Trends score
- Match Engagement = YouTube engagement metrics

3.5 Machine Learning Model

A Random Forest Regression model was used to predict future sponsorship value using historical IPL data.

Input Features:

- Popularity Score
- Match Engagement
- Attendance Proxy

Target Variable:

- Sponsorship Value Index (SVI)

The dataset was divided into:

- Training Data: 2021–2024
- Testing Data: 2025

The model performance was evaluated using:

- Mean Absolute Error (MAE)
- R^2 Score

3.6 Dashboard Development

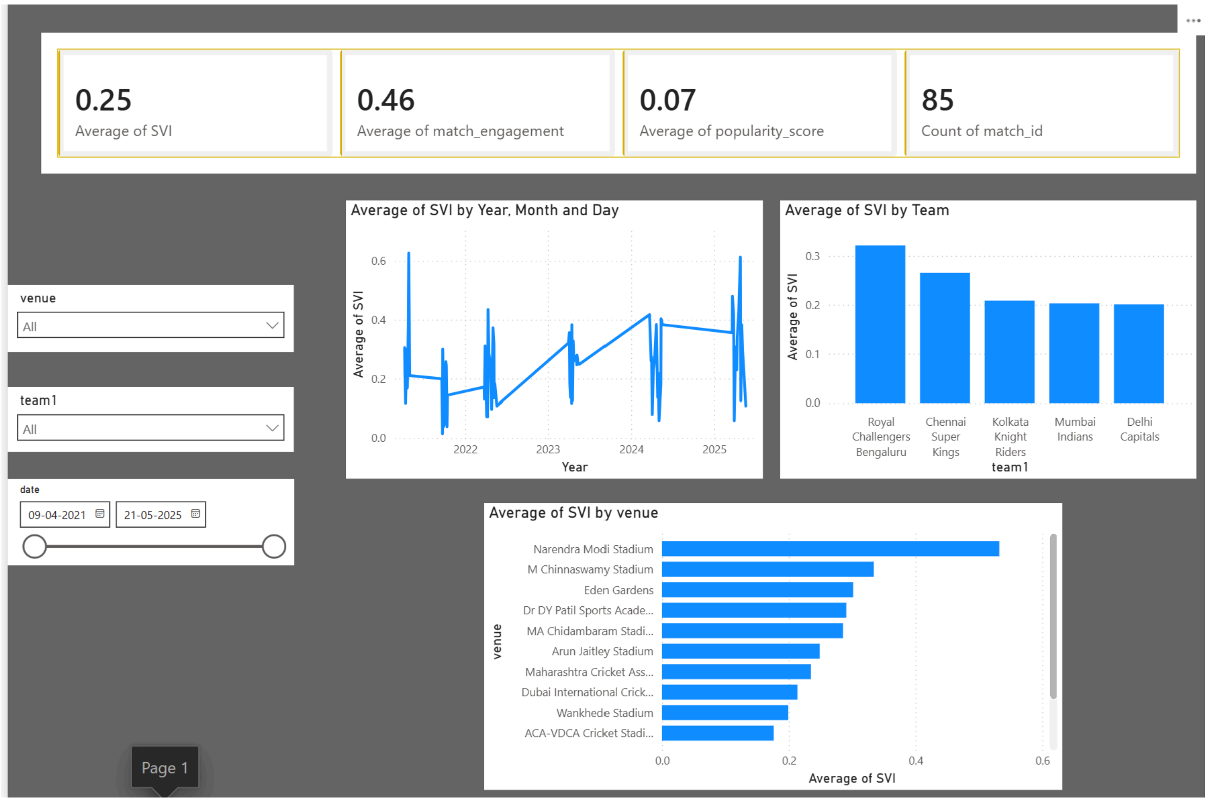
Power BI was used to develop an interactive dashboard consisting of:

1. Executive Overview
2. Fan Engagement Analysis
3. Predictive Sponsorship Analytics

The dashboard includes KPI cards, trend analysis, scatter plots, sponsorship rankings, and predictive visualizations.

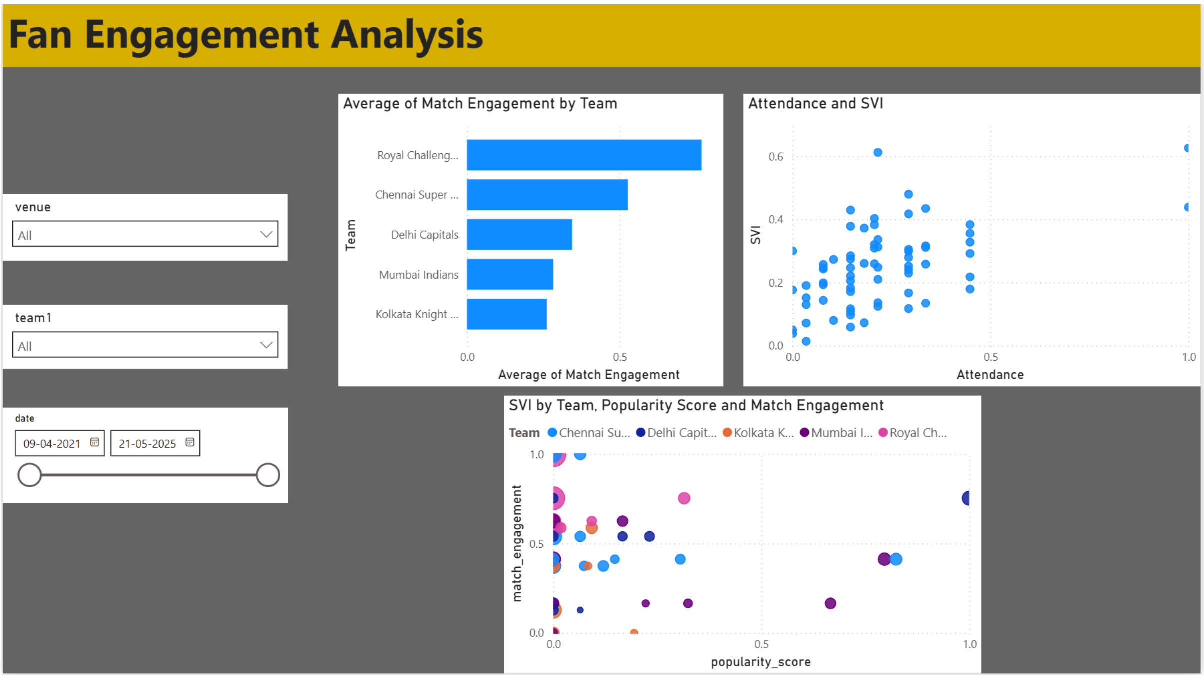
4. Results and Discussion

The developed dashboard successfully integrated IPL match data, social media engagement metrics, popularity analytics, and sponsorship intelligence into a unified analytical platform.



The Executive Overview dashboard highlighted sponsorship trends across IPL teams and venues. Results indicated that teams with higher fan engagement and popularity scores generally achieved higher Sponsorship Value Index (SVI) values.

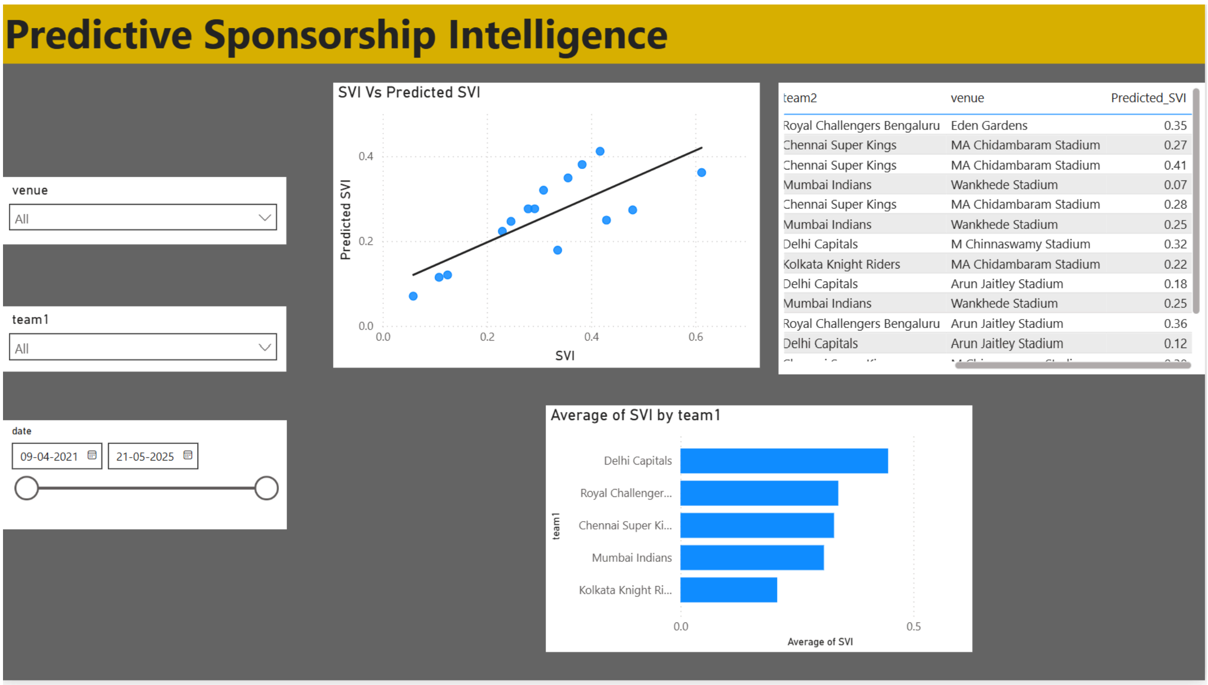
The Fan Engagement Analysis revealed that teams such as Royal Challengers Bengaluru and Chennai Super Kings demonstrated strong engagement levels across digital platforms. Scatter plot analysis showed a positive relationship between audience exposure, popularity, and sponsorship value.



Venue analysis indicated that larger stadiums such as Narendra Modi Stadium and M Chinnaswamy Stadium generated higher sponsorship value due to increased audience exposure.

The Machine Learning model demonstrated strong predictive performance in forecasting future sponsorship value using historical engagement and popularity patterns. The Actual vs Predicted SVI visualization showed a strong positive relationship between actual and predicted sponsorship values, indicating that the model effectively captured sponsorship trends.

The predictive analytics dashboard identified potential future sponsorship opportunities for IPL matches based on fan engagement and popularity indicators. The developed system demonstrates the effectiveness of integrating BI and ML techniques for sports sponsorship intelligence.



5. Conclusion

This project developed an IPL Sponsorship Value Intelligence Dashboard using Business Intelligence and Machine Learning techniques to analyze and predict sponsorship value across IPL matches and teams.

The study successfully integrated IPL match data, Google Trends popularity metrics, YouTube engagement analytics, and audience exposure indicators into a unified analytical framework. A Sponsorship Value Index (SVI) was developed to evaluate sponsorship opportunities using normalized engagement and popularity metrics.

The Power BI dashboard provided interactive insights into team performance, venue impact, fan engagement, and sponsorship trends. Additionally, the Machine Learning model demonstrated the capability to predict future sponsorship value using historical IPL engagement patterns.

The findings indicate that fan engagement, online popularity, and audience exposure significantly influence sponsorship value in modern sports leagues. The developed system can support sports organizations, sponsors, and analysts in making data-driven sponsorship decisions.

Future work may include integrating additional social media platforms such as Instagram and Twitter, incorporating real-time analytics, and expanding predictive modeling techniques for more advanced sponsorship forecasting.